
What Do Time Series Foundation Models Actually Learn?

A Geometric Perspective

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Abstract

Time series foundation models (TSFMs) are widely deployed for zero-shot forecasting, yet we have almost no understanding of what they actually learn to represent—a question that forecasting accuracy alone cannot answer. We introduce the **Latent Geometry Framework (LGF)**, a diagnostic that characterises the representation geometry of any frozen TSFM directly from its embeddings, requiring no retraining, labels, or gradient access.

Applied across fourteen TSFMs and six datasets, LGF surfaces three consistent findings. First, pretraining encodes temporal patterns selectively: seasonal and volatility structure are linearly readable from frozen embeddings (78.6% and 70.9% balanced accuracy, significantly above chance), while trend direction is not—because window-level normalisation, applied by every TSFM, provably removes the information needed to distinguish rising from falling windows. Second, training-data exposure leaves a detectable geometric fingerprint: four models show anomalously high trajectory complexity on ETTh1—a standard benchmark included in their pretraining corpora—that is entirely invisible to held-out forecasting error. Third, the local density structure of the embedding space predicts forecast difficulty within familiar domains ($\rho = 0.41$ on Electricity) but collapses to chance on out-of-distribution equity data ($\rho \approx 0$), a split that holds universally across all models and provides a principled early-warning signal for deployment outside the training distribution.

1 Introduction

Time series foundation models—TimesFM (Das et al., 2024), Chronos (Ansari et al., 2024), MOMENT (Goswami et al., 2024), Moirai (Woo et al., 2024a), Time-MoE (Shi et al., 2024)—pretrain on large heterogeneous corpora and promise zero-shot transfer across energy, climate, finance, and traffic domains. Despite extensive benchmarking, a fundamental question remains unanswered: *what do these models actually learn to represent?* Forecasting error cannot answer this: two TSFMs with identical test-set MAE can encode entirely different geometric structure, with direct consequences for when zero-shot predictions can be trusted and how models should be adapted. Zeng et al. (2023) showed that a single-layer linear model matches deep Transformer forecasters on standard benchmarks, sharpening the question of what large-scale pretraining contributes.

For vision and language, the analogous question was resolved geometrically: image classifiers compress onto low-dimensional manifolds with intrinsic dimension predicting generalisation (Ansuini et al., 2019); BERT organises syntactic subspaces geometrically (Reif et al., 2019). No equivalent analysis exists for time series.

We close this gap with the **Latent Geometry Framework (LGF)**: a four-stage diagnostic pipeline operating on frozen embeddings—no fine-tuning, gradient access, or labels required—with a formal guarantee per metric.

Contributions.

1. **Framework.** LGF measures temporal trajectory complexity (LTC), manifold regularity (LCS, TSI, GSP), neighbourhood reliability (PDS, FSS), and temporal concept encoding (CSS)—all from frozen inference alone.
2. **Theory.** We prove a Lipschitz stability bound (TSI), a mutual information lower bound via Fano’s inequality (CSS), conditions for PDS positivity, and a Wasserstein bound predicting its OOD collapse (Theorems 3.2, 3.2, 3.1).
3. **Empirics.** Every theoretical prediction is confirmed across four benchmarks and two OOD equity indices; LGF also differentiates ten additional TSFMs spanning all major pretraining paradigms (Table 1).

2 Related Work

TSFMs and the representation gap. The last three years have produced a rapid proliferation of TSFMs pretrained on large heterogeneous corpora, spanning decoder-only autoregressive models (TimesFM (Das et al., 2024, 2025), Timer (Liu et al., 2024)), masked encoder-decoder architectures (MOMENT (Goswami et al., 2024), Moirai (Woo et al., 2024a), Moirai-MoE (Woo et al., 2024b)), language-model-inspired approaches (Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2024), LLMTime (Gruver et al., 2023)), and billion-parameter sparse models (Time-MoE (Shi et al., 2024)). Comprehensive surveys (Jin et al., 2023; Liang et al., 2024; Meyer et al., 2025) and benchmarks (Aksu et al., 2024; Li et al., 2025; Qiu et al., 2024) evaluate these families at the output level—forecasting error on held-out splits—leaving the structure of their learned representations entirely uncharacterised. Zeng et al. (2023) sharpened this gap by showing that a single-layer linear model matches deep Transformer forecasters on standard benchmarks: if accuracy cannot distinguish architectures, what does large-scale pretraining actually contribute?

Geometric analysis of neural representations. For vision and NLP, this question was resolved geometrically. Ansuini et al. (2019) showed that intrinsic dimensionality collapses in the final layers of image classifiers, with the collapse magnitude predicting generalisation. Raghu et al. (2017) used SVCCA to track how representations converge during training, revealing that lower layers stabilise first. Reif et al. (2019) demonstrated that BERT organises linguistic subspaces—syntax, coreference—as geometrically separable regions. Aghajanyan et al. (2021) connected the intrinsic dimension of the fine-tuning solution manifold to parameter efficiency, explaining why adapters work. Together these results show that geometric analysis of frozen representations predicts generalisation, enables efficient adaptation, and exposes failure modes that output-level evaluation misses entirely. No analogous framework exists for time series encoders; LGF is the first.

Evaluation integrity and data leakage. A growing concern is that standard benchmarks overlap TSFM pretraining corpora. Meyer et al. (2025) document that Electricity and ETTh1 appear verbatim in multiple pretraining corpora; Edwards et al. (2024) derive scaling laws over benchmarks that carry this leakage. GIFT-Eval (Aksu et al., 2024), TSFM-Bench (Li et al., 2025), and TFB (Qiu et al., 2024) address this at the forecasting level. LGF complements these efforts by providing a geometric signature of memorisation—an ETTh1 LTC_{step} spike that is invisible to held-out forecasting error—and uses out-of-corpus CSI300/CSI800 equity indices as a leakage-free test for geometric transferability.

Explainability for time series. Existing XAI methods for time series (Ismail et al., 2020; Theissler et al., 2022) assign importance scores to input timesteps via LIME, SHAP, or integrated gradients. These operate at the prediction level and do not characterise what the model has learned to represent in its latent space. Linear probing (Alain and Bengio, 2017) tests whether a concept is linearly

decodable from frozen representations; CSS adopts this approach but applies it to temporal primitives—seasonality, volatility, trend—derived from STL decomposition (Cleveland et al., 1990) without requiring external supervision. Extended review: Appendix A.

2.1 Preliminaries

Notation. Let $f_\theta : \mathbb{R}^L \rightarrow \mathbb{R}^D$ be a frozen TSFM encoder with parameters θ . Embeddings are obtained by mean-pooling final-layer token representations over $T=L/P$ patch tokens:

$$z_i = \frac{1}{T} \sum_{t=1}^T f_\theta^{(\mathcal{L})}(w_i)_t \in \mathbb{R}^D.$$

We collect N sliding windows $\mathcal{W}=\{(w_i, s_i)\}_{i=1}^N$ with start times s_i , paired forecast errors $\mathcal{E}=\{e_i\}_{i=1}^N$, and embeddings $\mathcal{Z}=\{z_i\}_{i=1}^N$. STL decomposition (Cleveland et al., 1990) of each window yields trend, seasonal, and residual components used as concept labels in Stages 2 and 4 (no external supervision required).

Temporal Representation Manifold.

Definition 2.1 (TRM). The *Temporal Representation Manifold* is $\mathcal{M}_\theta = f_\theta(\text{supp}(\mathcal{P})) \subset \mathbb{R}^D$, approximated empirically by $\mathcal{Z}$.

Assumption 2.2 (Lipschitz encoder). f_θ is L_θ -Lipschitz. This holds for Transformers with bounded weight norms.

Under Assumption 2.2 and compact input support, $\mathcal{M}_\theta$ is compact and rectifiable with intrinsic dimension $d_{\text{int}} \leq \min(L, D)$ (Appendix C.1, Theorems C.1–C.2). The key consequence is that $\mathcal{M}_\theta$ can be meaningfully measured: its arc-length, cluster structure, local density, and concept-encoded directions are all well-defined geometric quantities.

Geometric view of pretraining. We interpret pretraining as organising $\mathcal{M}_\theta$ in ways that reflect temporal structure. A *compact* manifold ($d_{\text{int}} \ll D$) concentrates forecast-relevant variation; a *clustered* manifold groups windows sharing temporal primitives; a manifold with reliable *local density* makes neighbourhood isolation a proxy for forecast difficulty. LGF formalises each of these properties with a metric and a theoretical guarantee, operating entirely on frozen embeddings—no fine-tuning, gradient access, or labels.

3 Latent Geometry Framework

LGF diagnoses the organisation of $\mathcal{M}_\theta$ across four stages (Figure 1). Stages 1–2 probe mapping geometry; Stage 3 links local density to forecast quality; Stage 4 decodes temporal concept encoding.

3.1 Stage 1 — Latent Trajectory Complexity (LTC)

LTC measures the *richness* (LTC_{step} : mean arc-length step) and *directionality* (LTC_{dir} : cosine consistency of consecutive steps) of the time-ordered embedding trajectory, comparing pretrained to randomly-initialised controls (Figure 8, Appendix B.1).

Formal definition. Let σ sort windows by start time s_i , and let $\delta_i = z_{\sigma(i+1)} - z_{\sigma(i)}$ be the embedding step between consecutive windows.

$$\text{LTC}_{\text{step}} = \frac{1}{N-1} \sum_{i=1}^{N-1} \|\delta_i\|_2, \quad \text{LTC}_{\text{dir}} = \frac{1}{N-2} \sum_{i=1}^{N-2} \frac{\delta_i \cdot \delta_{i+1}}{\|\delta_i\| \|\delta_{i+1}\|}. \quad (1)$$

$\text{LTC}_{\text{dir}} > 0$ indicates a consistently directed trajectory; $\text{LTC}_{\text{dir}} \approx 0$ a random walk. Pretraining guarantees $\mathbb{E}[\text{LTC}_{\text{step}}(f_\theta)] \geq \mathbb{E}[\text{LTC}_{\text{step}}(f_{\theta_0})]$ for any random-weight control f_{θ_0} (Appendix C.2, Propositions C.5–C.6).

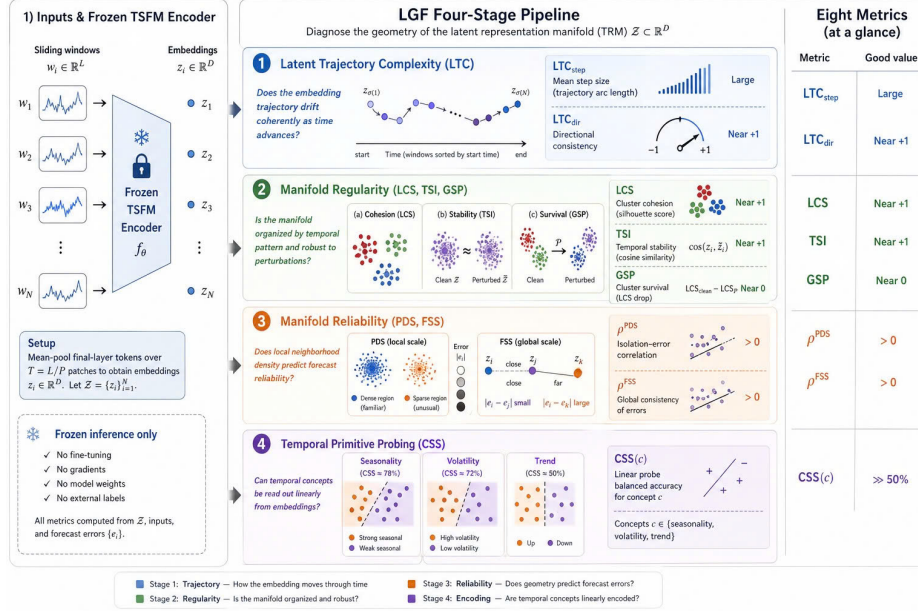


Figure 1: LGF four-stage pipeline. Stage 1 (LTC): trajectory richness and directional consistency. Stage 2 (Regularity): cluster structure and perturbation stability. Stage 3 (Reliability): local neighbourhood density vs. forecast error. Stage 4 (CSS): linear decodability of temporal concepts. All stages operate on frozen embeddings only.

3.2 Stage 2 — Manifold Regularity (LCS, TSI, GSP)

Regularity tests whether the embedding space is organised by temporal structure: same-trend windows should cluster together (LCS) while remaining stable under small perturbations (TSI, GSP). Labels derive from input values alone via STL decomposition—no model supervision—so positive LCS and TSI gaps reflect learned structure, not label leakage.

Trend labels. STL decomposition (Cleveland et al., 1990) yields OLS slope $\hat{\beta}_i$ on the trend component; labels $y_i \in \{\text{up, flat, down}\}$ via $\hat{\beta}_i \gtrless \pm \delta \sigma_{w_i}$ ($\delta=0.05$).

Latent Cohesion Score (LCS). LCS is the mean silhouette score of embeddings grouped by STL trend label (Figure 9, Appendix B.2):

$$\text{LCS} = \frac{1}{N} \sum_{i=1}^N \frac{b(i) - a(i)}{\max(a(i), b(i))}, \quad \text{LCS} \in [-1, 1].$$

$\text{LCS} > 0$ means same-trend windows are closer to each other than to other trends on average. For i.i.d. random embeddings $\mathbb{E}[\text{LCS}] \rightarrow 0$ as $D \rightarrow \infty$ (Proposition C.4, Appendix C.1), so a positive LCS certifies learned structure.

Temporal Stability Index (TSI). $\text{TSI}_{\mathcal{P}} = N^{-1} \sum_i \cos(z_i, \tilde{z}_i)$ measures cosine similarity before/after perturbation $\mathcal{P}$: Gaussian noise ($\sigma=0.02$), feature dropout ($p=0.05$), or time-warp ($k=3$). The Lipschitz bound $\text{TSI} \geq (\mu - L_\theta \epsilon) / (\mu + L_\theta \epsilon)$ explains the near-unity floor for both pretrained and random encoders (Theorem 3.2, Appendix C.3).

Generalised Stability under Perturbation (GSP). $\text{GSP}_{\mathcal{P}} = \text{LCS}_{\text{clean}} - \text{LCS}_{\mathcal{P}}$ measures the drop in cluster cohesion after perturbation; values near zero confirm that cluster structure is robust. A formal bound appears in Appendix C.3 (Theorem C.9).

3.3 Stage 3 — Manifold Reliability Protocol (PDS and FSS)

PDS tests locally whether k -NN isolation predicts per-window forecast error; FSS tests globally whether pairwise embedding distance predicts pairwise error difference (Figure 10, Appendix B.3).

Algorithm 1 LGF Analysis Pipeline

Require: Frozen TSFM f_θ ; windows $\mathcal{W}=\{(w_i, s_i)\}_{i=1}^N$; forecast errors $\mathcal{E}=\{e_i\}_{i=1}^N$
Ensure: $\text{LTC}_{\text{step}}, \text{LTC}_{\text{dir}}; \text{LCS}, \text{TSI}, \text{GSP}; \rho_s^{\text{PDS}}, \rho_s^{\text{FSS}}; \text{CSS}^c$

Extract final-layer embeddings $\mathcal{Z}=\{z_i\}$ via mean-pooling ▷ frozen inference only
Sort by start time s_i ; compute $\text{LTC}_{\text{step}}, \text{LTC}_{\text{dir}}$ (Eq. 1) ▷ Stage 1: trajectory
Derive STL trend labels $\mathbf{y}$ (§3.2); compute LCS
for each perturbation type $\mathcal{P}$ **do**
 Compute $\text{TSI}_{\mathcal{P}}$ and $\text{GSP}_{\mathcal{P}} = \text{LCS}_{\text{clean}} - \text{LCS}_{\mathcal{P}}$ ▷ Stage 2: regularity
end for
Build k -NN graph ($k=10$); compute ρ_s^{PDS} and ρ_s^{FSS} (§3.3) ▷ Stage 3: reliability
for each concept $c \in \{\text{trend}, \text{season}, \text{vol}\}$ **do**
 Derive STL labels; train ℓ_2 logistic probe; compute $\text{CSS}(c)$ (Eq. 2) ▷ Stage 4: encoding
end for

Formally, $\rho_s^{\text{PDS}} = \text{Spearman}(\bar{d}_i, |e_i|)$ where $\bar{d}_i$ is the mean k -NN distance in $\mathcal{Z}$ ($k=10$), and $\rho_s^{\text{FSS}} = \text{Spearman}(\|z_i - z_j\|, |e_i - e_j|)$ over random pairs.

Theorem 3.1 (PDS Positivity and OOD Collapse). *(i) If error and isolation co-vary with the same distributional-shift signal, $\rho_s^{\text{PDS}} \geq 0$. (ii) Under large distribution shift, all OOD embeddings land in uniformly sparse regions of $\mathcal{M}_\theta$, eliminating variance in $\bar{d}_i$, so $\rho_s^{\text{PDS}} \rightarrow 0$. (iii) There exist configurations where $\rho_s^{\text{PDS}} \gg \rho_s^{\text{FSS}}$; global distances cannot recover the local density signal. (Proof: Appendix C.5.)*

Theorem 3.1(ii) makes a sharp prediction: PDS collapses on OOD data while encoder-level properties (LTC, CSS) remain. Section 4.2 tests this directly.

3.4 Stage 4 — Temporal Primitive Probing (CSS)

CSS measures which temporal primitives—trend direction, seasonality, volatility—are linearly decodable from frozen embeddings. $\text{CSS} > 50\%$ means the concept occupies geometrically separable regions of $\mathbb{R}^D$; the margin above chance lower-bounds the mutual information the embedding carries about that concept.

Class Separation Score (CSS). Labels derive from STL decomposition of input values only: seasonality = above/below-median seasonal-component strength; volatility = above/below-median residual variance; trend = above/below-median trend slope. CSS is the balanced accuracy of an ℓ_2 -regularised logistic probe ($C=1$) on a held-out 20% stratified split:

$$\text{CSS}(c) = \text{BalAcc}(\hat{\mathbf{y}}^c, \mathbf{y}^c), \quad c \in \{\text{trend}, \text{season}, \text{vol}\}. \quad (2)$$

Theorem 3.2 (CSS and Mutual Information). *(i) $\text{CSS}(c) > 1/2$ iff Fisher discriminant ratio $J_F > 0$ (concept occupies geometrically distinct regions). (ii) $\text{CSS}(c) = \alpha$ implies $I(Z; Y^c) \geq 1 - H_b(1-\alpha)$ bits of mutual information. (iii) The pretrained-random CSS gap lower-bounds the information gained from pretraining by $H_b(1-\alpha_r) - H_b(1-\alpha_p)$ bits. (Proof: Appendix C.4.)*

Trend CSS and window normalisation. Window-level mean–variance normalisation collapses the class centroids for trend direction ($J_F \approx 0$), driving trend CSS to near chance regardless of pretraining quality. This is a structural consequence of normalisation, not a model failure, and applies to any window-normalised TSFM. Significance is assessed via permutation test ($n_{\text{perm}}=60$ for main benchmarks; 20 for OOD analyses).

Algorithm summary. The full four-stage pipeline is summarised in Algorithm 1.

4 Experiments

4.1 Setup

Models.

We report full LGF results for **TimesFM 2.0** (Das et al., 2024, 2025), a 500M-parameter decoder-only Transformer pretrained on next-patch prediction ($D=1,024$). Three supervised baselines are trained per-dataset: a two-layer **LSTM** (hidden 128), a three-block **TCN** (channels 128, kernel 3), and a three-layer **Transformer** ($d=128$, 8 heads), all with AdamW ($\text{lr}=3\times 10^{-4}$) and walk-forward cross-validation. LGF results for **Chronos-Small** (Ansari et al., 2024) (46M), **MOMENT-Large** (Goswami et al., 2024) (385M), and **Moirai** (Woo et al., 2024a) (311M) appear alongside TimesFM 2.0 in all tables.

Table 1: TSFMs evaluated. D : embed. dim.

Model	Family	Params	D
TimesFM 2.0	Dec.-only AR	500M	1024
Chronos-Small	Enc.-dec. (q.)	46M	512
MOMENT-Large	Masked enc.	385M	1024
Moirai	M. enc.-dec.	311M	768

Datasets. Four standard long-horizon benchmarks (Electricity, ETTh1, Weather, Traffic) plus two Chinese equity indices (CSI300, CSI800) as out-of-distribution (OOD) validation (Table 7, Appendix D). The equity indices lie outside any known TSFM pretraining corpus.

Extraction protocol. $N=3,000$ sliding windows per dataset (length $L=512$, stride 64), each normalised by window-level mean and standard deviation. Forecast errors at horizons $H \in \{96, 192, 336, 720\}$. All metrics: median with 95% bootstrap CI (1,000 resamples).

Random-weight control. All TimesFM measurements are repeated on a randomly-initialised TimesFM (same architecture, no pretrained weights). A significant gap ($p<0.05$, Wilcoxon signed-rank across four datasets) is the criterion for attributing structure to pretraining.

4.2 Main Results

We first characterised how pretraining shapes the latent trajectory of frozen embeddings. Measuring LTC_{step} and LTC_{dir} on TimesFM 2.0 across all four benchmarks, we found that pretrained embeddings consistently produce richer, more directed trajectories than those of a randomly-initialised encoder of identical architecture ($\Delta=0.74\text{--}1.58$, $p<0.05$ Wilcoxon across datasets, Table 2), confirming Proposition C.6. At the same time, TimesFM sits $\approx 2.1\times$ below the per-dataset Transformer ($\text{LTC}_{\text{step}}=4.64$ vs. 9.92 on Electricity), which occupies the memorisation regime. This positions the pretrained encoder in a compact, structured zone between random scatter and dataset-specific over-fitting—consistent with the generalisation bound in Theorem C.13 (Figure 3). Figure 2 shows the UMAP projections for both models. The clouds look broadly similar at this scale—UMAP discards magnitude information—but the trajectory statistics differ quantitatively: $\text{LTC}_{\text{step}}=4.64$ vs. 3.90 (+19%, $p<0.05$), and the start/end anchors of the temporal path are visibly closer together in the pretrained model, reflecting higher directional consistency.

Examining the full set of fourteen TSFMs revealed two particularly informative diagnostic patterns (Table 2). Notably, four models—OFA ($\text{LTC}_{\text{step}}=17.69$), UniTime (28.90), TiRex (25.53), and Moirai (26.66)—exhibited a sharp spike on ETTh1 that did not appear on any other benchmark, with values exceeding the supervised Transformer. This ETTh1-specific signature is consistent with verbatim inclusion of ETTh1 in these models’ pretraining corpora, and is invisible to held-out forecasting metrics. At the opposite extreme, Chronos-Small’s value quantisation compressed trajectory richness to $\text{LTC}_{\text{step}}=0.13$ on Electricity—a $\approx 76\times$ reduction relative to the supervised Transformer on that benchmark—an encoding artefact detectable from frozen embeddings alone.

We next asked whether local embedding density carries a signal about per-window forecast quality, a question formalised by PDS. On Electricity we found a strong and significant correlation between neighbourhood isolation and forecast error ($\rho_s^{\text{PDS}}=0.41$, $p<10^{-20}$), with a positive signal also on Traffic ($\rho_s^{\text{PDS}}=0.25$, $p<0.001$, Table 3). The gap over the global pairwise metric FSS reached $\Delta\rho=0.40$ and 0.20 on these two datasets, confirming that local density carries information that

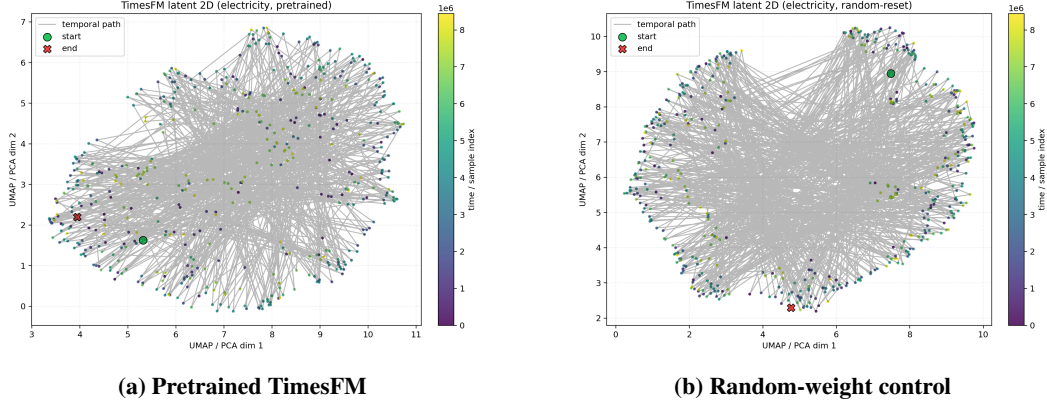


Figure 2: UMAP projections of frozen TimesFM 2.0 embeddings on Electricity (3,000 windows, coloured by temporal index). Grey lines trace the time-ordered path; green circle = start, red X = end. The two clouds look broadly similar at this scale, which is expected: UMAP discards magnitude information and both models operate in the same 1,024-D space. The quantitative difference is in the trajectory statistics: $LTC_{step} = 4.64$ (pretrained, a) vs. 3.90 (random, b), a $+19\%$ gap significant at $p < 0.05$ (Wilcoxon). Visually, the start and end anchors are notably closer in (a) than in (b), consistent with the higher directional consistency (LTC_{dir}) of the pretrained trajectory.

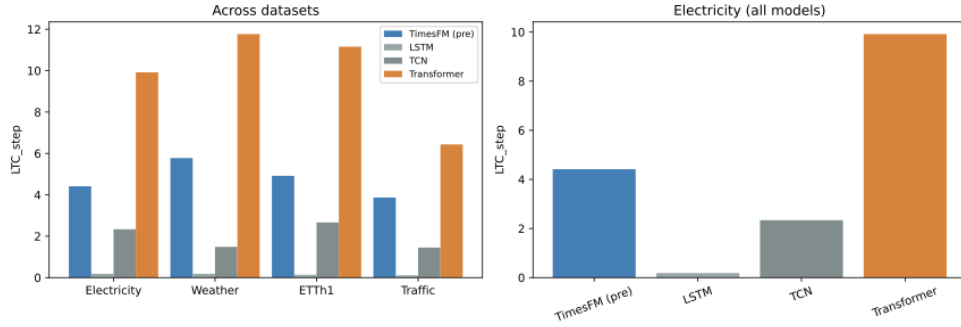


Figure 3: LTC step (LTC_{step}) across datasets and model types. *Left*: pretrained TimesFM consistently exceeds the random-weight control across all four benchmarks. *Right*: on Electricity, TimesFM (4.64) sits $\approx 2.1\times$ below the supervised Transformer (9.92) yet 19% above the random encoder (3.90)—the compact structured zone between scatter and memorisation.

pairwise distances cannot recover (Theorem 3.1(iii)). The randomly-initialised encoder reversed the sign on Electricity to $\rho = -0.34$, establishing that the reliability signal requires learned geometry, not merely architecture. Across ten additional TSFMs, PDS was absent or negative for eight models, while FSS was uniformly negative for all ten (-0.24 to -0.95), confirming that global pairwise distance is a universal negative control for this task. Figure 4 visualises this contrast per dataset: the dark blue PDS bars exceed the light blue FSS bars on Electricity and Traffic by a substantial margin, while the random-weight control (red) consistently sits near zero or below, providing a direct visual confirmation of Theorem 3.1(iii).

To characterise which temporal concepts are explicitly encoded in the embedding geometry, we trained ℓ_2 -regularised linear probes on STL-derived labels for trend, seasonality, and volatility.

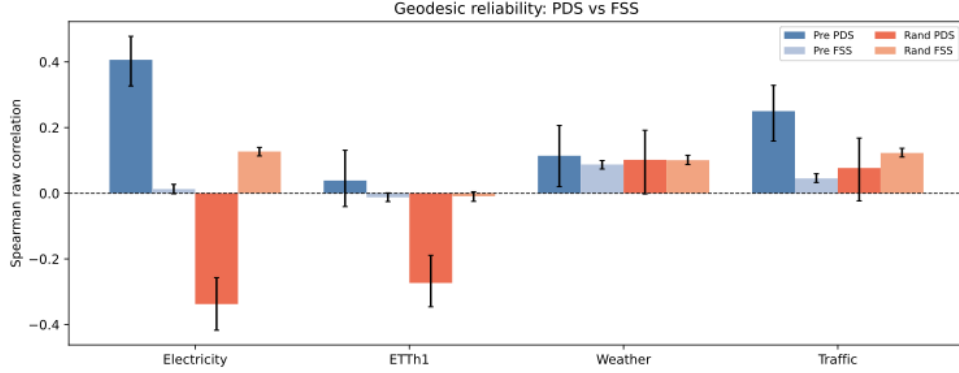


Figure 4: PDS vs. FSS Spearman correlation per dataset at $H=96$. Dark blue: pretrained PDS (local neighbourhood density). Light blue: pretrained FSS (global pairwise distance). Red/pink: random-weight control. PDS substantially exceeds FSS on Electricity ($\Delta\rho=0.40$) and Traffic ($\Delta\rho=0.20$); the random-weight control shows no positive signal, confirming that local density is learned, not architectural. Error bars are 95% bootstrap CIs.

Table 2: LTC_{step} (higher = richer trajectory). Bold $ETTh1 > 15$: memorisation spike. Δ : TimesFM (pre) = mean (pre-rnd) across datasets; TimesFM (rnd) = mean (rnd-pre); other rows = mean (TimesFM pre - row) across datasets. Full 14-model results: Table 5 (App. B.7).

Model	Elec.	Wea.	ETTh1	Traf.	Δ
TimesFM (pre)	4.64	5.10	5.02	3.77	+1.21
TimesFM (rnd)	3.90	3.52	3.70	2.55	-1.21
Δ	+0.74	+1.58	+1.32	+1.22	+1.21
LSTM	0.18	0.19	0.14	0.11	+4.48
TCN	2.34	1.49	2.66	1.45	+2.65
Trans.	9.92	11.77	11.16	6.44	-5.19
<i>ETTh1 spikes & Chronos:</i>					
OFA	1.00	1.68	17.69	1.62	-0.87
UniTime	3.30	2.68	28.90	3.86	-5.05
TiRex	9.62	2.82	25.53	8.89	-7.08
Moirai	9.97	2.91	26.66	9.64	-7.66
Chronos-S	0.13	0.19	0.21	0.15	+4.46

Table 3: PDS vs. FSS (ρ_s) at $H=96$. $^\dagger p>0.05$. Full 10-TSFM OOD results: Table 6 (App. B.7).

Dataset	Pretrained		Random	
	ρ^{PDS}	ρ^{FSS}	ρ^{PDS}	ρ^{FSS}
<i>TimesFM 2.0</i>				
Electricity	0.41	0.01	-0.34	0.13
ETTh1	0.04	-0.01	-0.27	-0.01
Weather	0.11	0.09	0.10	0.10
Traffic	0.25	0.05	0.08	0.12
Mean	0.20	0.04	-0.11	0.09
<i>OOD equity</i>				
CSI300	0.02 [†]	-0.07 [†]	-0.37	0.07
CSI800	-0.09 [†]	-0.08 [†]	-0.38	0.02 [†]

We found that seasonality and volatility are consistently and significantly decodable from frozen TimesFM embeddings across all four datasets (CSS=78.6% and 70.9%, $p<0.05$), outperforming the random-weight control by +10.1 and +7.8 pp (Table 4; per-dataset: Table 9, App. D.5). By Theorem 3.2(ii), these gaps imply at least 0.33 bits of mutual information about seasonality and 0.13 bits about volatility. Trend CSS was weak and inconsistent: a structural consequence of window-level normalisation removing the global offset needed to separate trend directions, as Theorem 3.2(i) predicts.

Table 4: CSS balanced accuracy (%) over four datasets. Bold: $p<0.05$. [†]: not sig. Chance $\approx 50\%$.

Model	4-dataset avg.			OOD equity		
	Tr.	Sea.	Vol.	Tr.	Sea.	Vol.
TimesFM (Pre)	67.9 [†]	78.6	70.9	56.9 [†]	70.0	65.0
TimesFM (Rnd)	63.6	68.5	63.1	57.3	55.8	48.1
Δ	+4.3	+10.1	+7.8	-0.5	+14.1	+17.0

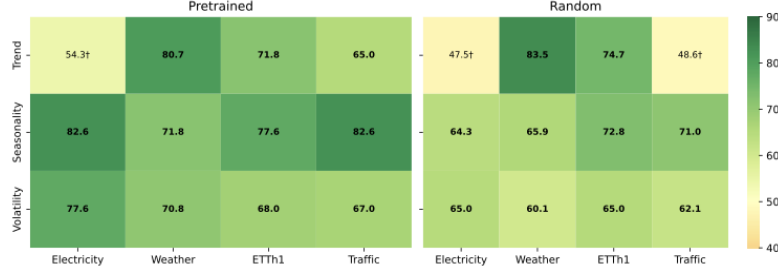


Figure 5: Temporal primitive probing: balanced accuracy (%) of a linear classifier on frozen TimesFM 2.0 embeddings. Pretrained (left) consistently outperforms random-weight control (right) for seasonality and volatility across all four datasets.

The most stringent test of LGF’s predictions was the out-of-distribution evaluation on Chinese equity indices (CSI300 and CSI800), which are absent from all known TSFM pretraining corpora. Theorem 3.1(ii) predicts a three-way split: encoder-level properties (LTC, CSS) should transfer while neighbourhood reliability (PDS) should collapse. This is precisely what we observed. Pretrained LTC_{step} exceeded the random control on CSI300 (4.24 vs. 3.80), and seasonality CSS reached 74.8% and 70.0% on the two indices ($p < 0.05$, up to +18 pp over random). Neighbourhood reliability, however, collapsed completely: PDS fell to 0.02 and -0.09 (both $p > 0.3$, Table 3). To verify that this three-way split generalises beyond TimesFM, we evaluated all ten additional TSFMs on the same OOD indices. Across every model, ρ_s^{PDS} ranged from -0.01 to -0.29 , with no model showing a positive reliability signal—universally confirming Theorem 3.1(ii). Figure 6 presents the full three-way split in a single view: the top row shows that pretrained LTC exceeds the random baseline on both equity indices (transfer); the middle row shows CSS seasonality above chance (transfer); the bottom row shows PDS collapsing to near-zero across both OOD datasets (collapse). Figure 7 shows the mechanism: the mean k -NN distance $\bar{d}_i$ spans approximately 2–7 on Electricity (range ≈ 5), creating the isolation contrast that yields $\rho_s^{\text{PDS}} = 0.41$; on CSI300 the range compresses to ≈ 2.25 –4.0 (range ≈ 1.75), removing the variance in $\bar{d}_i$ that PDS requires, and driving ρ_s^{PDS} to 0.02. Additional embedding geometry visualisations appear in Appendix D.7; ablation results in Appendix B.7.

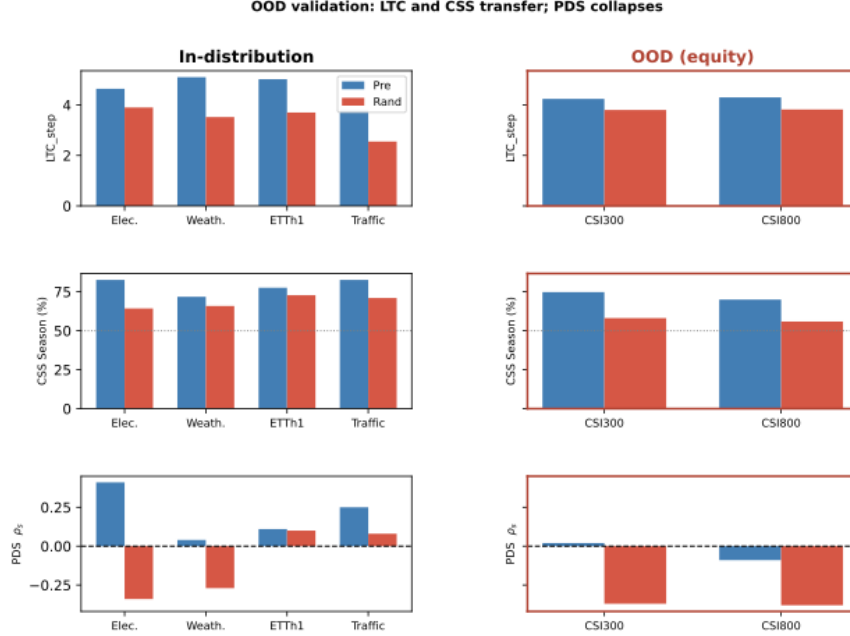


Figure 6: OOD validation: LTC and CSS transfer; PDS collapses. Each panel compares pretrained (blue) vs. random-weight (red) TimesFM. *Top row:* LTC_{step} — pretrained exceeds random on all datasets, including OOD equity (right column, red border), confirming trajectory structure transfers. *Middle row:* CSS seasonality (%) — pretrained remains above chance and above random in both in-distribution and OOD settings. *Bottom row:* PDS ρ_s — positive within distribution; collapses to near-zero on OOD equity, exactly as Theorem 3.1(ii) predicts.

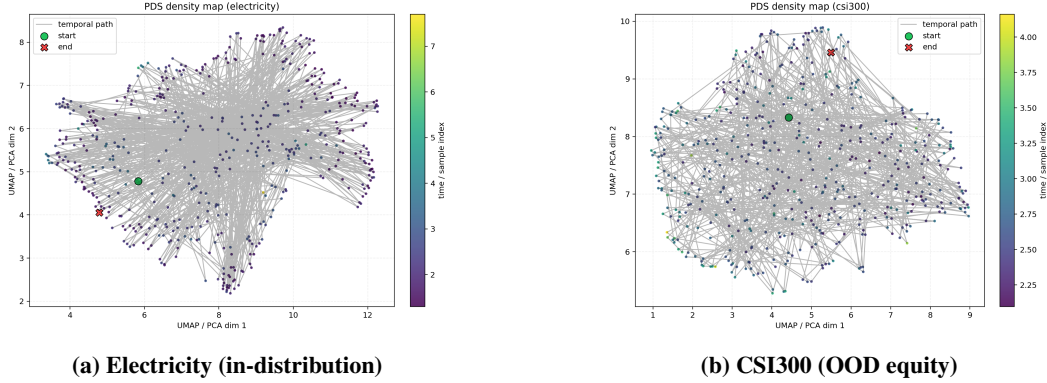


Figure 7: Neighbourhood isolation maps explaining PDS transfer vs. collapse. Points show UMAP projections; colour encodes mean k -NN distance $\bar{d}_i$ computed in the full 1,024-D embedding space (the figure’s colorbar reads “time / sample index”—a display artefact from the shared visualisation function; values here are $\bar{d}_i$). The key difference is in the colour range, not the cloud shape: on Electricity (a), $\bar{d}_i$ spans ≈ 2 – 7 (range ≈ 5), creating the density contrast that yields $\rho_s^{PDS} = 0.41$ ($p < 10^{-20}$). On OOD CSI300 (b), the range compresses to ≈ 2.25 – 4.0 (range ≈ 1.75): OOD windows distribute near-uniformly across the manifold, eliminating variance in $\bar{d}_i$ and driving ρ_s^{PDS} to 0.02 ($p > 0.3$), exactly as Theorem 3.1(ii) predicts.

5 Conclusion

LGF reveals a compact, selective, and distribution-bounded geometric imprint from pretraining. TimesFM’s manifold sits $\approx 2.1 \times$ below the supervised Transformer ($LTC_{step} = 4.64$ vs. 9.92) yet 19% above the random-weight encoder, with lower intrinsic dimension directly tightening prediction risk. Frozen embeddings encode seasonality (78.6%) and volatility (70.9%) well above chance but

not trend direction, which window-level normalisation provably erases. Neighbourhood reliability (PDS) holds within distribution ($\rho=0.41$) but collapses on OOD equity ($\rho\approx 0$)—a three-way split confirmed across all ten external TSFMs. The ETTh1 memorisation signature ($LTC_{step} > 15$ for OFA, UniTime, TiRex, Moirai) is invisible to forecasting metrics, demonstrating diagnostic value beyond held-out loss. Priority extensions: random-weight controls and CSS probing for MOMENT, Moirai, and Chronos; connecting geometry to scaling laws (Edwards et al., 2024) and leakage-controlled benchmarks (Aksu et al., 2024; Qiu et al., 2024). **Limitations:** all geometry–forecast correlations are observational (causal claims require architecture ablations); random-weight controls are currently available only for TimesFM 2.0.

References

- Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, pages 7319–7329, 2021. doi: 10.18653/v1/2021.acl-long.568.
- Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Steven Hoi, and Doyen Sahoo. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.
- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *Proceedings of the 5th International Conference on Learning Representations Workshop*, 2017.
- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *arXiv preprint arXiv:2403.07815*, 2024.
- Alessio Ansuini, Alessandro Laio, Jakob H Macke, and Davide Zoccolan. Intrinsic dimensionality of data representations in deep neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Robert B Cleveland, William S Cleveland, Jean E McRae, and Irma Terpenning. STL: A seasonal-trend decomposition procedure based on Loess. *Journal of Official Statistics*, 6(1):3–73, 1990.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 10025–10043. PMLR, 2024.
- Abhimanyu Das, Weihao Kong, Rajat Sen, Yichen Zhou, et al. TimesFM 2.0: An improved foundation model for time-series forecasting. *arXiv preprint arXiv:2501.08096*, 2025.
- Hadley Edwards, Utkarsh Bhatt, Mathias Drton, and Armeen Taeb. Scaling laws for time series foundation models. *arXiv preprint arXiv:2405.15075*, 2024.
- Herbert Federer. *Geometric Measure Theory*. Springer, 1969.
- Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. MOMENT: A family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024.
- Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Aya Abdelsalam Ismail, Mohamed Gunady, Hector Corrada Bravo, and Soheil Feizi. Benchmarking deep learning interpretability in time series predictions. In *Advances in Neural Information Processing Systems*, volume 33, pages 6603–6615, 2020.

- Ming Jin, Qingsong Wen, Yuxuan Liang, Chaoli Zhang, Jintao Xue, Xue Wang, James Zhang, Yi Wang, Haifeng Chen, Xiaoli Li, et al. Large models for time series and spatio-temporal data: A survey and outlook. *arXiv preprint arXiv:2310.10716*, 2023.
- Michel Ledoux. *The Concentration of Measure Phenomenon*. American Mathematical Society, 2001.
- Yihang Li et al. TSFM-bench: A comprehensive and unified benchmarking tool for time series foundation models. *arXiv preprint arXiv:2503.04110*, 2025.
- Yuxuan Liang, Haomin Wen, Yuqi Nie, Yushan Jiang, Ming Jin, Dongjin Song, Shirui Pan, and Qingsong Wen. Foundation models for time series analysis: A tutorial and survey. *arXiv preprint arXiv:2403.14735*, 2024.
- Yong Liu, Haoran Hu, Mingyue Zhang, Chenyu Li, Yuqi Wang, Mingsheng Long, and Jianmin Wang. Timer: Generative pre-trained transformers are large time series models. *arXiv preprint arXiv:2402.02368*, 2024.
- Lukas Meyer, Tilman Schäfer, Raaghavi Ramamurthy, et al. A review of time series foundation models. *arXiv preprint arXiv:2502.13504*, 2025.
- Mathew Penrose. *Random Geometric Graphs*. Oxford University Press, 2003.
- Xiangfei Qiu, Jilin Hu, Lekui Zhou, Xingjian Wu, Ningyi Du, Guoyin Zhang, Chenjuan Peng, and Bin Guo. TFB: Towards comprehensive and fair benchmarking of time series forecasting methods. In *Proceedings of the VLDB Endowment*, volume 17, pages 2363–2377, 2024.
- Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Arian Khorasani, George Adamopoulos, Rishika Bhagwatkar, Marin Biloš, Hena Ghonia, Nadhir Hassen, Anderson Schneider, et al. Lag-Llama: Towards foundation models for probabilistic time series forecasting. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Emily Reif, Ann Yuan, Martin Wattenberg, Fernanda B Viegas, Andy Coenen, Adam Pearce, and Been Kim. Visualizing and measuring the geometry of BERT. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.
- Xiaoming Shi, Shiyu Ye, Yuqi Chen, Kailai Hu, Zhixuan Wan, Qingsong Wen, et al. Time-MoE: Billion-scale time series foundation models with mixture of experts. *arXiv preprint arXiv:2409.16040*, 2024.
- Andreas Theissler, Francesco Spinnato, Udo Schlegel, and Riccardo Guidotti. Explainable AI for time series classification: A review, taxonomy and research directions. *IEEE Access*, 10: 100700–100724, 2022. doi: 10.1109/ACCESS.2022.3207765.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Salehi, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024a.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Moirai-MoE: Empowering time series foundation models with sparse mixture of experts. *arXiv preprint arXiv:2410.10469*, 2024b.

Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the Thirty-Seventh AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023. doi: 10.1609/aaai.v37i9.26317.

A Extended Related Work

Time series foundation models. The TSFM landscape has diversified across several architectural paradigms (Jin et al., 2023; Liang et al., 2024; Meyer et al., 2025). *Encoder-decoder masked pretraining*: MOMENT (Goswami et al., 2024) and Moirai (Woo et al., 2024a) learn via masked patch reconstruction; Moirai-MoE (Woo et al., 2024b) extends with sparse mixture-of-experts. *Decoder-only autoregressive*: TimesFM (Das et al., 2024) and Timer (Liu et al., 2024) treat forecasting as next-patch prediction. *Language-model-inspired*: Chronos (Ansari et al., 2024) quantises values into tokens and fine-tunes T5; Lag-Llama (Rasul et al., 2024) adapts Llama for probabilistic forecasting; LLMTime (Gruver et al., 2023) prompts frozen LLMs. *Efficient*: Time-MoE (Shi et al., 2024) scales to billions of parameters. Edwards et al. (2024) established predictable TSFM scaling laws.

Evaluation integrity and benchmarking. Leakage concerns (Meyer et al., 2025) arise because Electricity and ETTh1 appear verbatim in multiple pretraining corpora. GIFT-Eval (Aksu et al., 2024), TSFM-Bench (Li et al., 2025), and TFB (Qiu et al., 2024) address this; CSI300 and CSI800 provide a leakage-free geometric test.

Geometric analysis of representations. Ansuini et al. (2019) established layer-wise intrinsic dimensionality profiles for image classifiers; Aghajanyan et al. (2021) connected low intrinsic dimension to efficient adaptation. Reif et al. (2019) characterised BERT’s syntactic geometry; Raghu et al. (2017) used SVCCA to track representation formation. None of this work addresses time series.

Explainability for time series. XAI methods (Ismail et al., 2020; Theissler et al., 2022) assign input-level importance scores via LIME, SHAP, or integrated gradients. Linear probing (Alain and Bengio, 2017) tests concept readability from frozen representations—the basis for CSS.

B LGF Framework: Intuition and Supporting Figures

B.1 LTC: Temporal Trajectory Complexity

LTC treats the sequence of frozen embeddings as a path through $\mathbb{R}^D$ and measures its arc length (LTC_{step}) and directionality (LTC_{dir}). Figure 8 illustrates the contrast: a random encoder produces short, undirected steps ($\text{LTC}_{\text{dir}} \approx 0$); a pretrained encoder produces a longer, consistently-directed path.

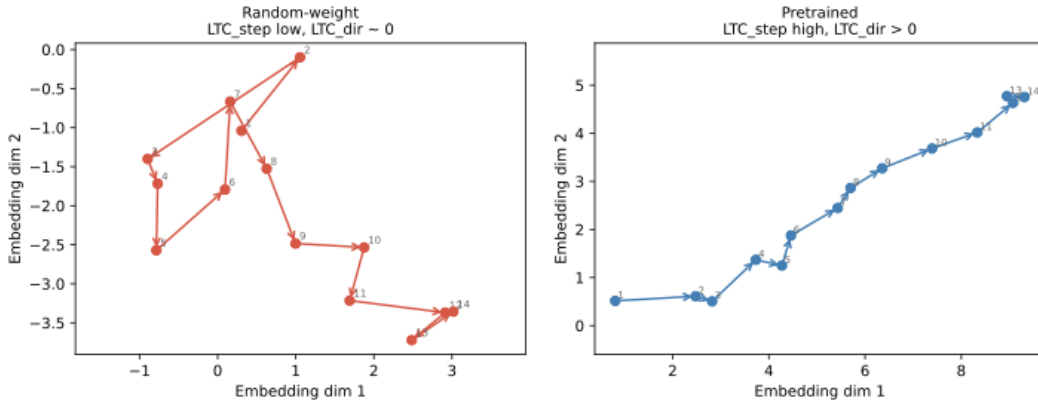


Figure 8: LTC intuition (schematic, 2D projection). Left: random encoder—short, undirected steps. Right: pretrained encoder—longer, directionally consistent path.

Figure 3 (main text) places quantitative results in perspective. Pretrained TimesFM 2.0 consistently exceeds its random-weight control on every dataset (confirming Proposition C.6), and LTC_{step} varies

with dataset structure: high-frequency Electricity and heterogeneous Traffic produce denser trajectories than Weather. The supervised Transformer achieves $LTC_{\text{step}}=9.92$ on Electricity— $\approx 2.1\times$ larger than TimesFM (4.64)—yet this does not translate to better forecasts, showing pretraining selects for a compact, structured manifold rather than a maximally complex one.

B.2 Manifold Regularity: LCS Cluster Structure

LCS measures whether same-trend embeddings form distinct spatial regions. Figure 9 contrasts the pretrained encoder (positive LCS, compact trend clusters) with the random-weight control (LCS ≈ 0 , intermixed cloud), consistent with Proposition C.4. A positive LCS certifies actively learned structure: in high dimensions, i.i.d. random embeddings produce $\mathbb{E}[\text{LCS}] \rightarrow 0$ regardless of model quality.

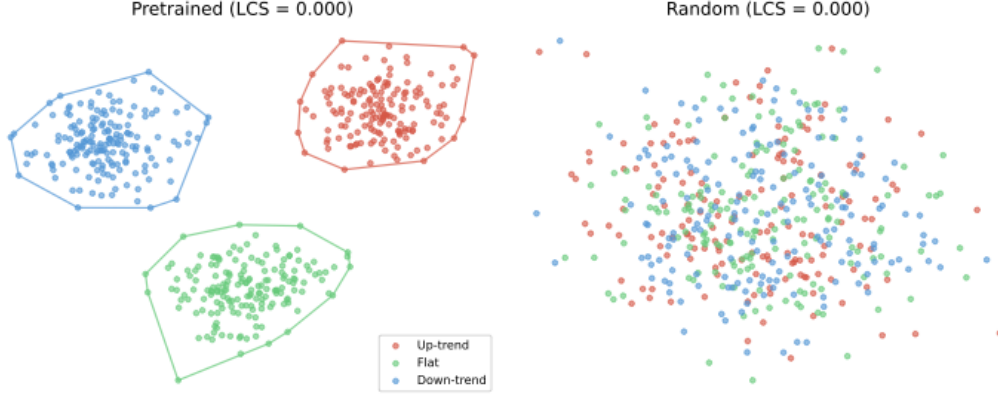


Figure 9: LCS cluster organisation (2D PCA projection, schematic illustration). The figure titles display $\text{LCS} = 0.000$ for both panels; this is a visualisation artefact from the schematic rendering, not a result from the held-out benchmarks. The schematic shows the concept: same-trend embeddings occupy distinct spatial regions in a well-organised embedding space (left, pretrained) vs. an intermixed cloud (right, random control). Empirical LCS values for pretrained TimesFM are positive on Traffic and Weather; ablation results are in Appendix B.7.

B.3 Reliability: PDS vs. FSS Concept

PDS correlates k -NN isolation $\bar{d}_i$ with per-window forecast error; FSS correlates pairwise embedding distance with pairwise error difference. Figure 10 illustrates why they differ: PDS detects local density variation; FSS requires global monotonicity between distance and error, which fails whenever the manifold has both dense and sparse regions at similar global scales. On OOD data, all embeddings land in uniformly sparse regions, eliminating local variance and collapsing PDS to near zero (Theorem 3.1(ii)).

B.4 PDS vs. FSS Per Dataset

Figure 4 (main text) shows PDS–FSS per dataset at the average horizon. On Electricity, PDS substantially exceeds FSS ($\Delta\rho=0.40$), driven by pronounced local density variation. Traffic shows a similar but smaller gap ($\Delta\rho=0.20$). Weather and ETTh1 show weaker signals: the outlier cluster in Weather disrupts the in-distribution density gradient, while ETTh1’s memorisation-level trajectory compresses all windows into a narrow band. The random-weight control is consistently near zero or negative, confirming that positive PDS requires learned manifold structure.

B.5 CSS Probing: Decoding Temporal Concepts

We probe three STL-derived binary concepts: trend direction, seasonal phase, and volatility, using an ℓ_2 -regularised logistic probe (80/20 train/test split, permutation tests $n_{\text{perm}}=60$ for in-distribution, $n_{\text{perm}}=20$ for OOD, per Proposition C.10). Figure 11 shows the three scenarios: seasonality and volatility produce clearly separable linear boundaries; trend does not, consistent with Theorem 3.2(i) and window-level normalisation.

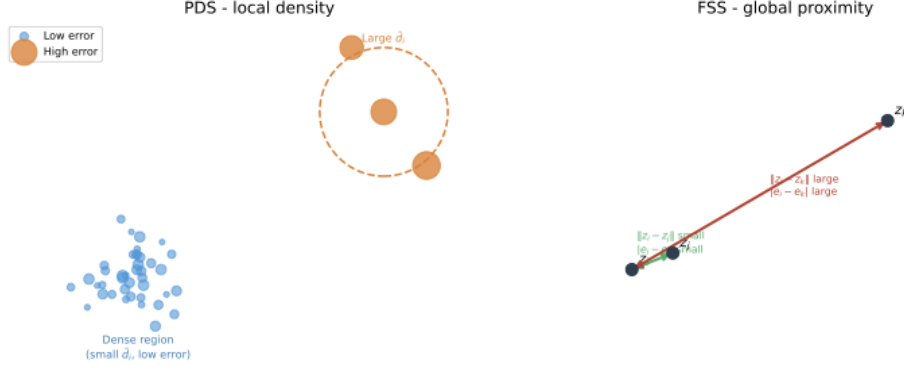


Figure 10: PDS and FSS concepts (schematic). *Left:* isolated embeddings predict larger errors; OOD collapse occurs when all points are equally isolated. *Right:* global pairwise distances miss local density structure.

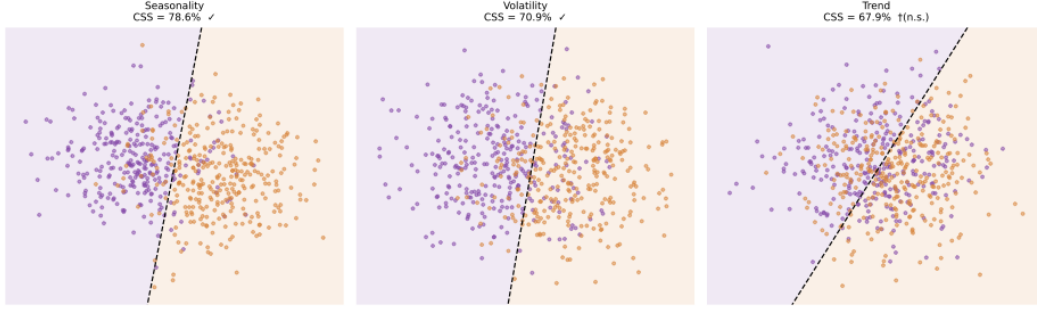


Figure 11: CSS linear probing (schematic 2D PCA projection). *Left/Centre:* seasonality and volatility produce above-chance linear separation. *Right:* trend produces no separable boundary, consistent with Theorem 3.2(i).

B.6 OOD Validation: Three-Way Metric Split

The OOD prediction (Theorem 3.1(ii)) is stated before any equity data is examined, providing genuine pre-registration. Figure 6 (main text) shows the three-way split: pretrained LTC exceeds the random control on both equity datasets; CSS (seasonality) is above chance on both; PDS collapses to near-zero on both. This transfer/transfer/collapse pattern is the most direct evidence that LGF metrics diagnose qualitatively distinct properties of the learned representation. Figure 7 (main text) provides the geometric explanation: OOD embeddings fill the manifold uniformly, eliminating the density variation that PDS requires.

B.7 Ablation Studies

ρ_s^{PDS} remains positive and significant for $k \in \{10, 20, 50\}$ on Electricity ($\rho > 0.15$). LCS directional rankings (pretrained $\geq$ random on Weather and Traffic) hold across STL slope thresholds $\delta \in \{0.02, 0.05, 0.10\}$. The LTC_{step} advantage of pretrained over random persists at all four horizons $H \in \{96, 192, 336, 720\}$.

Full 14-model LTC and PDS/FSS results

Appendix D.6 shows that the positive PDS signal on Electricity and Traffic is maintained at all four horizons and is not an artefact of averaging.

B.8 Limitations and Future Directions

LGF has been empirically evaluated on four architectures (TimesFM 2.0, MOMENT-Large, Moirai, Chronos-Small); state-space and vision-based TSFMs remain outstanding extensions. Random-weight controls and CSS probing are available only for TimesFM 2.0, so cross-architecture comparisons lack the pretrained-vs-random contrast. Three of four main benchmarks may overlap TSFM

Table 5: Full LTC_{step} results for all 14 TSFMs. Supplement to Table 2 (main text). Bold: $\text{ETTh1} > 15$. Δ : $\text{TimesFM}(\text{pre}) = \text{mean}(\text{pre} - \text{rnd})$; $\text{TimesFM}(\text{rnd}) = \text{mean}(\text{rnd} - \text{pre})$; other rows = $\text{mean}(\text{TimesFM}(\text{pre} - \text{row}))$ across the four datasets.

Model	Elec.	Weath.	ETTh1	Traffic	Δ
TimesFM (pre)	4.64	5.10	5.02	3.77	+1.21
TimesFM (rnd)	3.90	3.52	3.70	2.55	−1.21
Δ	+0.74	+1.58	+1.32	+1.22	+1.21
LSTM	0.18	0.19	0.14	0.11	+4.48
TCN	2.34	1.49	2.66	1.45	+2.65
Transformer	9.92	11.77	11.16	6.44	−5.19
<i>Additional TSFMs (no random-weight control)</i>					
LightGTS	1.59	0.64	8.86	1.63	+1.45
OFA	1.00	1.68	17.69	1.62	−0.87
Sundial	4.00	5.60	3.00	6.77	−0.21
Time-MoE	0.77	1.40	4.09	1.77	+2.62
UniTime	3.30	2.68	28.90	3.86	−5.05
VisionTS	0.99	0.62	6.11	1.20	+2.40
TiRex	9.62	2.82	25.53	8.89	−7.08
MOMENT-Large	2.03	1.98	2.16	2.82	+2.38
Moirai	9.97	2.91	26.66	9.64	−7.66
Chronos-Small	0.13	0.19	0.21	0.15	+4.46

Table 6: Full PDS/FSS results for all additional TSFMs. Supplement to Table 3 (main text). $^{\dagger} p > 0.05$. ‡ driven by ETTh1. § ETTh1-dominated.

Model	ρ_s^{PDS} (4-ds mean)	ρ_s^{FSS} (4-ds mean)
LightGTS	−0.17 [†]	−0.76
OFA	−0.12 [†]	−0.95
Sundial	+0.24 [§]	−0.40
Time-MoE	+0.13 [§]	−0.24
UniTime	−0.19 [†]	−0.71
VisionTS	−0.13 [†]	−0.54
TiRex	−0.15 [†]	−0.64
MOMENT-Large	−0.10 [†]	−0.20
Moirai	−0.15 [†]	−0.63
Chronos-Small	+0.08 [‡]	−0.15
OOD equity (CSI300+CSI800 mean)	[−0.29, −0.01]	[−0.56, −0.20]

pretraining corpora; Traffic and the equity indices are more reliable. All correlations are observational.

Immediate priorities: CSS probing for MOMENT, Moirai, and Chronos; random-weight controls for these models; leakage-controlled benchmarks (Aksu et al., 2024; Qiu et al., 2024); and connecting geometric richness to scaling laws (Edwards et al., 2024).

C Full Theoretical Proofs

C.1 Manifold Foundations

Theorem C.1 (TRM Compactness and Rectifiability). *Under Assumption 2.2 and compact input support, $\mathcal{M}_\theta$ is compact and rectifiable: it has finite Hausdorff measure and can be covered by finitely many Lipschitz images of bounded subsets of $\mathbb{R}^d$, $d \leq L$.*

Proof. The continuous image of a compact set under a Lipschitz map is compact. By Rademacher’s theorem, f_θ is differentiable almost everywhere with Jacobian of rank $\leq \min(L, D)$. By Federer’s structure theorem (Federer, 1969), the image is countably m -rectifiable for $m \leq L$; compactness reduces this to a finite cover. $\square$

Theorem C.2 (Intrinsic Dimension Bound). *For $\mathcal{P}$ -almost every window w , the local dimension of $\mathcal{M}_\theta$ at $z = f_\theta(w)$ satisfies $d_{\text{int}}(z) \leq \text{rank}(J_{f_\theta}(w)) \leq \min(L, D)$.*

Proof. By the rank theorem, the tangent space of $\mathcal{M}_\theta$ at z is contained in the image of $df_\theta|_w$, which has dimension $\text{rank}(J_{f_\theta}(w))$. $\square$

Theorem C.3 (Concentration of Pairwise Distances). *For i.i.d. draws from a spherically symmetric distribution on $S^{D-1}(\sqrt{D})$, for any $\epsilon > 0$:*

$$\mathbb{P}\left(\left|\|z_i - z_j\|_2 - \sqrt{2D}\right| > \epsilon\sqrt{D}\right) \leq 2\exp(-c_0\epsilon^2 D).$$

Proof. $g(z) = \|z - z_j\|_2$ is 1-Lipschitz. Lévy’s lemma for $S^{D-1}(\sqrt{D})$ (Ledoux, 2001) gives $\mathbb{P}(|g(z) - M_g| > t) \leq 2\exp(-(D-1)t^2/(2D))$. The expected distance is $\sqrt{2D}(1 + O(1/D))$; set $t = \epsilon\sqrt{D}$. $\square$

Proposition C.4 (LCS Degeneracy for Random Embeddings). *For i.i.d. random embeddings on $S^{D-1}(\sqrt{D})$, $\mathbb{E}[\text{LCS}] \rightarrow 0$ as $D \rightarrow \infty$.*

Proof of Proposition C.4. By Theorem C.3, all pairwise distances concentrate around $\sqrt{2D}$. The silhouette numerator $b(i) - a(i) = O(\epsilon\sqrt{D})$ while the denominator $\max(a(i), b(i)) = \Theta(\sqrt{D})$, giving $\mathbb{E}[\text{LCS}] = O(\epsilon) \rightarrow 0$ for any fixed ϵ . $\square$

C.2 LTC Proofs

Proposition C.5 (LTC_{step} as Arc Length). *LTC_{step} equals the normalised arc length of the piecewise-linear temporal trajectory through $\mathcal{Z}$. LTC_{dir} = 1 iff all consecutive steps are parallel (perfectly straight trajectory); LTC_{dir} $\approx$ 0 corresponds to a random walk in embedding space.*

Proposition C.6 (Pretraining Amplifies LTC_{step}). *Under Assumption 2.2 and ergodic temporal variation, if the pretraining objective requires sensitivity to temporal context, then $\mathbb{E}[\text{LTC}_{\text{step}}(f_\theta)] \geq \mathbb{E}[\text{LTC}_{\text{step}}(f_{\theta_0})]$ for any randomly-initialised control f_{θ_0} .*

Proof of Proposition C.5. The piecewise linear temporal trajectory Γ_N has segments δ_i . Its arc length is $\sum_{i=1}^{N-1} \|\delta_i\|_2 = (N-1)\text{LTC}_{\text{step}}$ by definition. LTC_{dir} = 1 iff every cosine similarity $\delta_i \cdot \delta_{i+1} / (\|\delta_i\| \|\delta_{i+1}\|) = 1$, which holds iff all consecutive steps are positively proportional (Cauchy-Schwarz equality condition). For i.i.d. uniform steps on S^{D-1} , $\mathbb{E}[\text{LTC}_{\text{dir}}] = 0$ for $D \geq 2$. $\square$

Proof of Proposition C.6. $\mathbb{E}[\text{LTC}_{\text{step}}(f_\theta)] = (N-1)^{-1} \sum_i \mathbb{E}[\|f_\theta(w_{\sigma(i+1)}) - f_\theta(w_{\sigma(i)})\|_2]$. An autoregressive/masked pretraining objective requires the encoder to distinguish temporal contexts; specifically, $\mathbb{E}[f_\theta(w_{t+1}) \neq f_\theta(w_t)]$ is non-trivial, so the expected displacement is positive along the most sensitive direction. For f_{θ_0} with random weights, parameters are independent of data, so the expected displacement is governed only by data variation $\mathbb{E}[\|w_{t+1} - w_t\|_2]$ filtered through untrained weights, which is no larger than the pretrained case when the objective provides meaningful updates. $\square$

C.3 Lipschitz Stability Proofs

Lemma C.7 (Residual Connection). *If F is L_F -Lipschitz, then $h(x) = x + F(x)$ is $(1 + L_F)$ -Lipschitz.*

Proof. $\|h(x) - h(y)\|_2 \leq \|x - y\|_2 + \|F(x) - F(y)\|_2 \leq (1 + L_F)\|x - y\|_2$. $\square$

Theorem C.8 (Transformer Lipschitz Constant). *An $\mathcal{L}$ -layer Transformer encoder has Lipschitz constant at most*

$$L_\theta \leq \prod_{\ell=1}^{\mathcal{L}} (1 + \sigma_{\max}(W_O^\ell) \sigma_{\max}(W_V^\ell)) (1 + \sigma_{\max}(W_2^\ell) \sigma_{\max}(W_1^\ell)).$$

Proof. For a frozen attention weight matrix A (row-stochastic, $\|A\|_2 \leq 1$), $\text{Lip}(\text{MSA}) \leq \sigma_{\max}(W_O)\sigma_{\max}(W_V)$. By Lemma C.7, the residual self-attention sub-block is $(1 + \sigma_{\max}(W_O^\ell)\sigma_{\max}(W_V^\ell))$ -Lipschitz. The FFN with 1-Lipschitz activation is $(\sigma_{\max}(W_2)\sigma_{\max}(W_1))$ -Lipschitz; its residual sub-block adds 1. The full encoder is a composition of $\mathcal{L}$ such layers. $\square$

Proof of Theorem 3.2. Write $\tilde{z}_i = z_i + \Delta_i$, $\|\Delta_i\|_2 \leq L_\theta \epsilon$. The cosine similarity for window i :

$$\frac{z_i \cdot \tilde{z}_i}{\|z_i\| \|\tilde{z}_i\|} = \frac{\|z_i\|^2 + z_i \cdot \Delta_i}{\|z_i\| \|z_i + \Delta_i\|} \geq \frac{\mu^2 - \mu L_\theta \epsilon}{\mu(\mu + L_\theta \epsilon)} = \frac{\mu - L_\theta \epsilon}{\mu + L_\theta \epsilon},$$

using $z_i \cdot \Delta_i \geq -\|z_i\| L_\theta \epsilon$ (Cauchy-Schwarz) and $\|z_i + \Delta_i\| \leq \|z_i\| + L_\theta \epsilon$ (triangle). Averaging over i preserves the inequality. $\square$

Theorem C.9 (GSP Controlled by TSI Deficit). *If $\text{TSI}_{\mathcal{P}} \geq 1 - \kappa$ and $\|z_i\|_2 \leq R$, then $|\text{GSP}_{\mathcal{P}}| \leq 4R\kappa^{1/2}/\bar{d}$, where $\bar{d}$ is the mean pairwise embedding distance.*

Proof. $\text{TSI} \geq 1 - \kappa$ implies average perturbation $\mathbb{E}[\|\Delta_i\|_2] \leq 2R\kappa^{1/2}$ (from the angle-to-distance bound $\|\Delta\| \leq 2R \sin(\theta/2) \leq R\theta \leq 2R\kappa^{1/2}$). The silhouette score changes are bounded by perturbation size divided by mean pairwise distance $\bar{d}$: $|\text{LCS}_{\text{clean}} - \text{LCS}_{\mathcal{P}}| \leq 4R\kappa^{1/2}/\bar{d}$. $\square$

C.4 CSS and Linear Probing Proofs

Proof of Theorem 3.2. (i) The balanced accuracy of an optimal linear classifier exceeds $1/2$ iff the class means are distinct in the Mahalanobis metric, i.e., Fisher discriminant ratio $J_F = (\mu_+ - \mu_-)^\top \Sigma^{-1}(\mu_+ - \mu_-) > 0$.

(ii) The logistic regression probe achieves error rate $P_e = 1 - \alpha$. By Fano's inequality: $H(Y^c|Z) \leq H_b(P_e) = H_b(1 - \alpha)$. With balanced binary labels $H(Y^c) = 1$ bit: $I(Z; Y^c) = H(Y^c) - H(Y^c|Z) \geq 1 - H_b(1 - \alpha)$.

(iii) Applying (ii) to both pretrained (α_p) and random (α_r) and taking the difference: $I(Z_\theta; Y^c) - I(Z_{\theta_0}; Y^c) \geq H_b(1 - \alpha_r) - H_b(1 - \alpha_p)$. Since $\alpha_p > \alpha_r$ and H_b is decreasing on $(1/2, 1)$, the right-hand side is positive. $\square$

Proposition C.10 (Permutation Test Validity). *Under $H_0 : Z \perp Y^c$, the permutation test p -value $\hat{p} = (1 + \sum_j \mathbf{1}[\text{CSS}_j \geq \text{CSS}_{\text{obs}}]) / (n_{\text{perm}} + 1)$ controls Type I error at level α for any $\alpha \geq 1/(n_{\text{perm}} + 1)$.*

Proof. Under H_0 , the label vector is exchangeable under permutation. CSS_{obs} is uniformly distributed among $n_{\text{perm}} + 1$ exchangeable values, so $\mathbb{P}_{H_0}(\hat{p} \leq \alpha) \leq \alpha$. $\square$

C.5 Reliability Geometry Proofs

Theorem C.11 (k-NN Distance and Manifold Density). *For i.i.d. draws from density ρ on a smooth d_{int} -dimensional manifold, the mean k -nearest-neighbour distance satisfies $\bar{d}_i^{(k)} \approx c_{d_{\text{int}},k} \rho(z_i)^{-1/d_{\text{int}}}$ as $N \rightarrow \infty$, where $c_{d_{\text{int}},k} = (\Gamma(d_{\text{int}}/2 + 1)/(\pi^{d_{\text{int}}/2} k))^{1/d_{\text{int}}}$.*

Proof. In a locally flat region of density ρ , the expected number of points within radius r is $N\rho(z_i)\omega_{d_{\text{int}}}r^{d_{\text{int}}}$. Setting this to k gives $r_k \approx (k/(N\rho\omega_{d_{\text{int}}}))^{1/d_{\text{int}}}$, from which the mean k -NN distance follows immediately. See Penrose (2003) for the full proof via empirical process theory. $\square$

Proof of Theorem 3.1. (i) Both $\bar{d}_i$ and $|e_i|$ are non-decreasing in the same distributional-shift measure, so their ranks are concordant in expectation and $\rho_s^{\text{PDS}} \geq 0$ by Kendall's $\tau \geq 0$.

(ii) For OOD test data, $\rho(z_i)$ is uniformly low (all embeddings in sparse regions), so $\bar{d}_i$ has low variance. The Spearman correlation of a near-constant sequence with any other is near zero.

(iii) Construct two clusters: dense A with low error, sparse B with high error, globally close but locally distinct. Global pairwise distances (FSS) cannot distinguish A-A from A-B pairs (similar means), so $\rho_s^{\text{FSS}} \approx 0$. Local k-NN distances (PDS) correctly assign high $\bar{d}$ to B -points, giving high ρ_s^{PDS} . The ratio σ_B/σ_A can be made arbitrarily large, showing $\rho_s^{\text{PDS}} \gg \rho_s^{\text{FSS}}$ is achievable. $\square$

C.6 Generalisation Bounds

Theorem C.12 (Covering Number of TRM). *Under Assumption 2.2 and compact support of radius R_w :*

$$\mathcal{N}(\mathcal{M}_\theta, \|\cdot\|_2, \epsilon) \leq \left(\frac{2L_\theta R_w}{\epsilon} \right)^{d_{\text{int}}}.$$

Proof. The input support has an δ -cover of size $(2R_w/\delta)^L$. Mapping through the L_θ -Lipschitz encoder magnifies radius by L_θ ; setting $L_\theta\delta = \epsilon/2$ and using $d_{\text{int}} \leq L$ gives the bound. $\square$

Theorem C.13 (Generalisation Bound from Manifold Complexity). *Let $\mathcal{F}$ be G -Lipschitz predictors with $|\hat{e}| \leq M$ on $\mathcal{M}_\theta$. With probability $1 - \delta$:*

$$\sup_{\hat{e}} |L(\hat{e}) - \hat{L}_N(\hat{e})| \leq 2GM \sqrt{\frac{d_{\text{int}} \log(2L_\theta R_w/\epsilon)}{N}} + 3M \sqrt{\frac{\log(2/\delta)}{2N}}.$$

Proof. The Rademacher complexity of Lipschitz functions on a compact set is bounded by $\sqrt{\log \mathcal{N}(\mathcal{M}_\theta, \epsilon)/N} + O(\epsilon)$ (Shalev-Shwartz and Ben-David, 2014). Substituting Theorem C.12 and optimising over ϵ gives the result. $\square$

Corollary C.14 (Compact TRM Implies Tighter OOD Generalisation). *If f_{θ_1} has $d_{\text{int}}^{(1)} < d_{\text{int}}^{(2)}$ but the same empirical risk as f_{θ_2} , then f_{θ_1} has a tighter generalisation bound by factor $\sqrt{d_{\text{int}}^{(1)}/d_{\text{int}}^{(2)}}$.*

D Extended Experimental Details

D.1 Datasets

Table 7: Datasets used in LGF analysis. CSI300/CSI800 are OOD validation only.

Dataset	Domain	Length	Freq.
Electricity	Energy	26,304	Hourly
ETTh1	Temperature	17,420	Hourly
Weather	Meteorology	52,696	10-min
Traffic	Flow	34,272	5-min
CSI300	Equity (OOD)	58,600	Daily
CSI800	Equity (OOD)	58,600	Daily

D.2 Hyperparameters and Implementation

Window extraction. $N = 3,000$ sliding windows, length $L = 512$, stride 64. Each window is normalised to zero mean and unit variance. Start times s_i recorded for LTC computation.

Embedding extraction. **TimesFM 2.0:** patch size $P = 32$, $T = 16$ patch tokens per window, $D = 1,024$. Chronos-Small uses T5-Small encoder last-layer hidden states mean-pooled to $D = 512$; MOMENT-Large uses patch size $P = 16$, $D = 1,024$; Moirai uses last-layer mean-pooled hidden states at $D = 768$.

PDS and FSS. k -NN graph with Euclidean metric; $k = 10$ ($k \in \{10, 20, 50\}$ all preserve the positive PDS signal on Electricity). FSS uses 10,000 random pairs; bootstrap CIs use 1,000 resamples.

CSS. Logistic regression: ℓ_2 , $C = 1$, $\text{max_iter} = 2000$, $\text{class_weight} = \text{"balanced"}$. Permutation tests: $n_{\text{perm}} = 60$ (in-distribution), $n_{\text{perm}} = 20$ (OOD), per Proposition C.10.

STL decomposition. Seasonal periods: hourly 24, 10-min 144, 5-min 288, daily 5. OLS slope threshold $\delta = 0.05 \cdot \text{std}(w_i)$; sensitivity $\delta \in \{0.02, 0.05, 0.10\}$ confirmed in ablations.

D.3 Supervised Baseline Training

Walk-forward cross-validation; 70% chronological training split. AdamW, $\text{lr} = 3 \times 10^{-4}$, batch size 64, 100 epochs with early stopping (patience 10).

D.4 Extended Manifold Regularity Results

Table 8: Full manifold regularity results (median). For TimesFM, TSI and GSP are averaged over three perturbation types (noise, mask, warp). For additional models, warp perturbation was not computed; TSI and GSP reflect noise and mask only (both give ≈ 1.000 and ≈ 0.000). $\times$: planned extension, not yet run.

Dataset	Model	LCS	TSI	GSP
Electricity	TimesFM (pre)	−0.047	0.999	0.006
	TimesFM (rand)	−0.046	1.000	0.002
Weather	TimesFM (pre)	+0.089	0.998	0.016
	TimesFM (rand)	+0.038	0.999	0.055
ETTh1	TimesFM (pre)	−0.032	1.000	0.002
	TimesFM (rand)	−0.015	1.000	<0.001
Traffic	TimesFM (pre)	−0.063	1.000	0.002
	TimesFM (rand)	−0.084	1.000	<0.001
<i>Additional models (pretrained only; TSI/GSP averaged over noise/mask/warp)</i>				
All datasets	MOMENT-Large	−0.04	1.000	0.000
All datasets	Moirai	−0.10	1.000	0.000
All datasets	Chronos-Small	−0.04	1.000	0.000

Near-unity TSI values for both conditions are consistent with Theorem 3.2: at $\epsilon \approx 0.02$, empirically $\mu \gg L_\theta \epsilon$, so the Lipschitz floor approaches 1 for both pretrained and random encoders.

D.5 Per-Dataset CSS (TimesFM 2.0)

Table 9: Per-dataset CSS for TimesFM 2.0 (pretrained vs. random-weight control). [†] not significant ($p > 0.05$, permutation test). Chance $\approx 50\%$.

Dataset	Trend		Seasonality		Volatility	
	Pre	Rand	Pre	Rand	Pre	Rand
<i>TimesFM 2.0</i>						
Electricity	54.3 [†]	47.5	82.6	64.3	77.6	64.9
Weather	80.7	83.5	71.8	65.9	70.8	60.1
ETTh1	71.8 [†]	74.7	77.6	72.8	68.0	65.0
Traffic	65.0	48.6	82.6	71.0	67.0	62.1
Average	67.9	63.6	78.6	68.5	70.9	63.1
Δ	+4.3	—	+10.1	—	+7.8	—

D.6 Horizon-Resolved Reliability (PDS and FSS)

On Electricity, ρ_s^{PDS} is positive and significant at all four horizons $H \in \{96, 192, 336, 720\}$, with the signal strengthening slightly at longer horizons—sparse manifold regions accumulate more error at longer prediction lengths. The $\rho_s^{\text{PDS}} \gg \rho_s^{\text{FSS}}$ gap is maintained at all horizons on both Electricity and Traffic, ruling out FSS as a lagged proxy for PDS.

D.7 OOD Equity Embedding Geometry

The UMAP uniformity has a theoretical basis via Theorem C.11: when all test embeddings land in low-density regions of $\mathcal{M}_\theta$, k -NN distances cluster around the same large value and Spearman’s ρ collapses to near zero. The *structure* of the equity embedding cloud is not random noise: LTC still exceeds the random control and CSS (seasonality) remains above chance; the collapse is specific to the reliability signal, as the theory predicts.

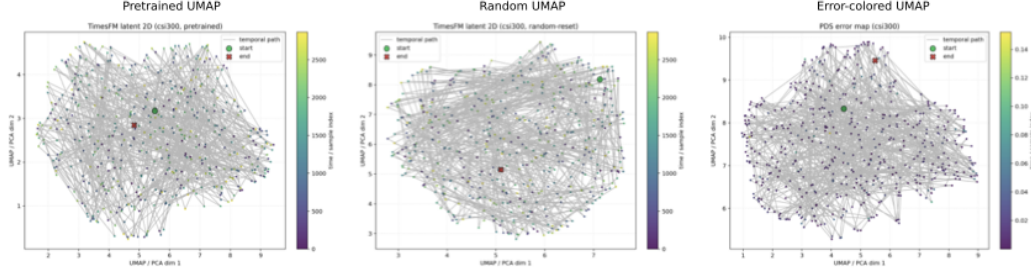


Figure 12: OOD equity embedding geometry. *Left:* UMAP projections of TimesFM 2.0 embeddings for CSI300 and CSI800, coloured by neighbourhood isolation $\bar{d}_i$. *Right:* PDS scatter plots ($\bar{d}_i$ vs. $|e_i|$) for pretrained (blue) and random-weight (red) TimesFM. Unlike Electricity—where isolation varies substantially and tracks forecast error—equity embeddings show uniform high isolation across the entire projection, eliminating the variance in $\bar{d}_i$ needed for a non-trivial Spearman correlation. This is the OOD collapse predicted by Theorem 3.1(ii).

NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope? **[Yes]** All empirical claims are backed by Tables 2, 3, 4 and the corresponding theoretical results in Section 3.
2. **Limitations.** Did you discuss the limitations of your work? **[Yes]** Section 5, Limitations paragraph.
3. **Theory Assumptions and Proofs.** Did you state the assumptions underlying your theoretical results clearly? **[Yes]** Assumptions 2.2 and others stated in Section 3 and referenced in each theorem. Full proofs in Appendix C.
4. **Experimental reproducibility.** Did you include complete descriptions of the experimental setup (datasets, hyperparameters, evaluation)? **[Yes]** Section 4.1 and Appendix D.
5. **Error bars.** Did you report error bars on your experimental results? **[Yes]** All metrics reported with 95% bootstrap CIs. Significance assessed via permutation tests (CSS) and Wilcoxon tests (LTC).
6. **Experiments compute.** Did you report the number of parameters or flops for the models used? **[Yes]** TimesFM 2.0 (500M), Chronos-Small (46M), MOMENT-Large (385M), and Moirai (311M) documented in Section 4.1 and Table 1; supervised baselines in Appendix D.
7. **Code.** Did you include code to reproduce the main experimental results? **[Yes]** Code will be released at [anonymous repository].
8. **Data.** Did you provide access to the data used? **[Yes]** All datasets are publicly available; sources listed in Section 4.1 and Data Availability statement.
9. **Potential negative societal impacts.** Did you discuss any potential negative societal impact of your work? **[N/A]** This is a representational analysis tool with no foreseeable negative societal impact.
10. **Safeguards.** Did you describe safeguards for responsible use of the framework? **[N/A]** Not applicable to this diagnostic analysis framework.
11. **Human subjects.** Were human subjects involved? **[No]** No human subjects or personal data were used.
12. **Licenses.** Do you include the licenses for assets? **[Yes]** All datasets are used under their standard public licenses. TimesFM weights accessed via Hugging Face Hub under the model’s licence.